



Agricultural inputs, urbanization, and urban-rural income disparity: Evidence from China

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ABSTRACT

Income inequality is undoubtedly a potential economic and social risk for any country in the world. For China, the improvement in agricultural production capacity and the steady progress of urbanization are fundamental guarantees to mitigate unbalanced urban-rural development and alleviate urban-rural conflict. However, previous studies have paid little attention to the effects of agricultural inputs such as chemical fertilizers on urban-rural income disparity. Based on a data set of 30 provincial-level regions in China over 1997–2015, we use the system generalized method of moments to investigate the impacts of agricultural production inputs and urbanization on urban-rural income disparity. The results show that increasing urbanization has a significant effect on mitigating urban-rural income disparity. The intensity of chemical fertilizer application also has a significant impact on the disparity, but the impact depends on the level of urbanization. For the provinces with relatively low levels of urbanization, an increase in the intensity of chemical fertilizer application can moderate the disparity, while a decline in the intensity of chemical fertilizer application can narrow the disparity in more urbanized provinces. The threshold levels of urbanization present a time-varying characteristic; however, the threshold effects are significant over the entire sample period. Therefore, during the process of further urbanization in China, it is necessary and urgent to appropriately adjust the pattern of chemical fertilizer application and thus reduce the over-dependence on chemical fertilizers in agricultural production.

1. Introduction

The socioeconomic problems caused by income inequality can result in some potential risks of low-efficient economic development (Alesina & Perotti, 1996; Alesina & Rodrik, 1994; De La Croix & Doepke, 2003; Galor & Zeira, 1993). Although in recent years, the urban-rural relationship in China has been improved to some extent, the urban-rural dual structure is still very obvious and the urban-rural income disparity remains high (Chen & Lin, 2013; Xie & Zhou, 2014), curbing the long-term growth of China's economy and the improvement in the efficiency of resource allocation (Wang & Ouyang, 2008; Yang, 1999; Yang & Zhou, 1999). Hence, China needs to effectively alleviate the harms that the urban-rural income disparity may cause to economic efficiency and social equity. Under such a background, it is very necessary to explore how to mitigate urban-rural income disparity and thus provide policy references for reducing the potential risks of economic development and enhancing the stability of long-term economic growth in China.

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At the end of 2016, China's population was 1.3827 billion and the urbanization rate was 57.35%.¹ In the same year, there were 134.921 million ha of arable land in China,² but the arable land per capita was only 0.0976 ha, which was almost half of 0.192 ha—the world's average.³ These specific national conditions imply that the discrepancy between grain demand and supply in China will remain in the long term, and the prominence of the discrepancy between population and land will increase with the acceleration of urbanization. Once agricultural production encounters negative shocks, the process of urbanization and the coordinated development of urban and rural areas will be hindered (Christiansen, 2009; Gollin, Parente, & Rogerson, 2002; Zhang & Wan, 2015). In the face of such pressures, to increase grain yields and secure food supply, China's agricultural production has become increasingly dependent on agrochemicals, such as chemical fertilizers.

The characteristics of high input and high output of China's agricultural production are clear. From 1978 to 2016, China's grain production doubled, and the total amount of agricultural fertilizer application increased over 5.77 times.⁴ At present, China applies the greatest chemical fertilizers in the world, accounting for approximately one third of the global total. In terms of intensity, China's agricultural fertilizer application is 328.5 kg per ha, which is substantially higher than the world average of 120 kg per ha (the Ministry of Agriculture of the People's Republic of China, 2015).

The main cause for the staggering amount of chemical fertilizer application in China over the past half century is the yield-increasing effect of chemical fertilizers on agricultural production (Heisey & Norton, 2007). Lin (1992) argued that the contribution of the household contract responsibility system (HRS) to the total value of agricultural output over the period 1978–1984 was 46.89%, and the contribution of chemical fertilizer application was 32.20%. After the HRS was fully promoted in China, the contribution of chemical fertilizer application over the period 1984–1987 was as high as 53.71%. Wang and Xiao (2008) found that the contribution of chemical fertilizer application to grain yields over the period 1978–2006 was the greatest among all agricultural inputs, which was just < 57%. While the population-land relationship is particularly tense in China's rural areas, the increase in grain yield caused by chemical fertilizer application implies that agricultural productivity has increased. This is generally regarded as one of the main reasons for the release of rural surplus labor forces from their lands in China (Lin, 1992; Wang & Xiao, 2008).

The significant yield-increasing effect of chemical fertilizer application is related to China's rural revitalization, helps maintain reasonable and stable agricultural product prices, and provides guarantees for the coordination of the urban-rural relationship. However, thus far, the mechanisms regarding how the application of chemical fertilizers affects the incomes of urban and rural households remain unclear. Does the application of chemical fertilizers mitigate urban-rural income disparity by directly improving agricultural production conditions and thus increasing farmers' agricultural income, or by promoting urban-rural migration and thus increasing rural households' wage income? Moreover, what impacts will urbanization have on the application of chemical fertilizers and urban-rural income disparity?

Although some of the literature has focused on examining the role of urbanization in narrowing urban-rural income disparity (Chen & Lin, 2013; Lu & Chen, 2006; Zhong, 2011), Du and Liu (2009) found that the increase in the level of urbanization would significantly reduce the total amount of chemical fertilizer and pesticide application. They argued that the income-increasing effect of urbanization reduces the dependence of agricultural production on agrochemicals. Therefore, if the application of chemical fertilizers plays an important role in mitigating urban-rural income disparity, the indirect effect of urbanization mentioned above should not be ignored. In addition, although some studies have examined the impacts of inputs, such as agrochemicals and agricultural machinery, on farmers' incomes (Du & Liu, 2009; Van den Berg et al., 2007), few have directly assessed the impacts of agricultural production inputs on urban-rural income disparity. Using a panel data set of 30 provincial-level regions in China over the period 1997–2015, this paper conducts a novel empirical examination of the contributions of the intensity of chemical fertilizer application, urbanization, and their interaction regarding the mitigation of urban-rural income disparity.

The widespread application of agrochemicals in agricultural production is conducive to increasing agricultural output and represents an important basis for urbanization. However, the long-term overuse of agrochemicals has caused incalculable damages to environmental quality, agricultural ecology, grain securities, and human health (Savci, 2012; Zhu & Chen, 2002). Taking these serious consequences into consideration, and to effectively advance the supply-side reforms in fertilizer industry, the Ministry of Agriculture of the People's Republic of China (MOA) issued the *Action Plan of Zero Growth in Fertilizer Use* and the *Action Plan of Zero Growth in Pesticide Use* in 2015. These two zero-growth programs put strict limits on the amount of chemical fertilizer and pesticide application in China. On the one hand, if the application of chemical fertilizers helps farmers increase their income and thus mitigates urban-rural income disparity, policymakers should actively explore other means to narrow urban-rural income disparity while the zero-growth program aims to reduce the excessive dependence of agricultural production on chemical fertilizers. On the other hand, and more important, is whether the restrictions imposed by the MOA on the application of chemical fertilizers will have policy impacts on the growth of farmers' incomes and urban-rural relationships. To accurately assess the policy impacts of the zero-growth targets, it is necessary to examine the direct impact of chemical fertilizer application on urban-rural income disparity.

The remainder of the paper is organized as follows. Section 2 presents the literature review and theoretical mechanism. The

¹ Source: *Statistical Bulletin of the People's Republic of China on National Economic and Social Development in 2016* released by the National Bureau of Statistics of China.

² Source: *Statistical Bulletin of the People's Republic of China on Land, Mineral and Marine Resources 2017* released by the Ministry of Natural Resources of the People's Republic of China.

³ Source: World Bank Data, <https://data.worldbank.org/indicator/AG.LND.ARBL.HA.PC?end=2016&start=1961> (accessed on February 23, 2019).

⁴ Source: authors' calculation based on the online database of the National Bureau of Statistics of China.

empirical methodology, variables, and data are detailed in Section 3. Section 4 offers the empirical results on the impacts of chemical fertilizer application, urbanization, and the interaction between these two variables on urban-rural income disparity. Section 5 further identifies the threshold effects of chemical fertilizer application on narrowing the urban-rural income gap and the impacts of other agricultural inputs, such as pesticides and agricultural plastic films, on the urban-rural income gap. Section 6 presents the concluding remarks.

2. Literature review and theoretical mechanism

In the process of urbanization, the urban-rural conflicts involving property rights, income, migration, and education impair resource allocation efficiency and social equity between urban and rural areas (Ann, Wu, Zheng, Zhang, & Shen, 2014). Among these risks, the continuously expanding urban-rural income disparity is a main obstacle to advance the integration of urban and rural development and eliminate the urban-rural dual structure in China (Knight & Song, 1999; Lin & Chen, 2011; Lu & Chen, 2006). The literature has shown there are several factors which have widened urban-rural income disparity, namely, government's prioritized development strategy of heavy industries or capital-intensive sectors (Chen & Lin, 2013; Kanbur & Zhang, 2005; Lin & Chen, 2011), the household registration system and *hukou*-based discrimination (Liu, 2005; Lu & Chen, 2006), financial development (Chen, Liu, & Zhang, 2010), fiscal expenditures on education (Sicular, Yue, Gustafsson, & Li, 2007), and other urban-biased institutions and policies (Yang, 1999). By contrast, the urbanization process helps narrow urban-rural income disparity (Chen & Lin, 2013; Lu & Chen, 2006).

After China's reform and opening up in 1978, the HRS reform contributed toward narrowing the urban-rural income disparity in China by encouraging rural households to increase their agricultural output (Chen et al., 2010; McMillan, Whalley, & Zhu, 1989; Yang & Zhou, 1999). However, after the HRS reform was completed in the late 1980s, the economic incentives rendered by the reform gradually faded. Since the late 1980s, township and village enterprises have played an important role in narrowing urban-rural income disparity by providing rural surplus labor forces with a number of non-agricultural jobs. However, this effect declined annually and was eventually exhausted after 1998, with the continuous increase in the competitiveness of urban industrial enterprises (Zhong, 2011). Regarding governmental financial support, Shen and Zhang (2007) found that even rural-biased fiscal expenditures on rural public infrastructure had little effect on narrowing urban-rural income disparity.

The role of agricultural inputs (including chemical fertilizers) in mitigating urban-rural income disparities has received less attention. The first step of investigating this potential effect is to explain why the large-scale application of chemical fertilizers is often observed in developing countries, especially in China. Heisey and Norton (2007) indicated that the scarcity of arable land has been an important cause of the dramatic increase in the demand for chemical fertilizers in developing countries and high-yield seeds and declining fertilizer prices driven by technological progress in the fertilizer industry have been an impetus for the increase in the use of chemical fertilizers in agricultural production.

The government's intervention in fertilizer prices is another main cause of the increased use of chemical fertilizers in China. Since the reform and opening up, although China's decision-making mechanism for fertilizer production and prices has gradually shifted from the government plan to the market, the Chinese government has long maintained the practice of subsidizing the fertilizer industry to reduce production costs and prices. According to Li (2014), the central government's total subsidies for fertilizer production and application increased from 7.89 billion *yuan* in 2003 to 157 billion *yuan* in 2011, of which subsidies for fertilizer production increased from 7.89 billion *yuan* to 63.1 billion *yuan*. Li (2014) also found that the central government's subsidies for the production and use of effective components of chemical fertilizers increased from 200 *yuan* per ton to 2480 *yuan* per ton. To protect farmers' incomes and ensure grain security, the government also adopted some packages of comprehensive subsidies for agricultural production to reduce the costs of chemical fertilizers. As a result, the actual expenditures for farmers to purchase chemical fertilizers are lower than the costs of chemical fertilizers, leading to an increase in chemical fertilizer application.

The large-scale application of chemical fertilizers has significantly increased grain yields in China (Lin, 1992; Wang & Xiao, 2008). This is of great importance for understanding urban-rural income disparity. The use of chemical fertilizers helps increase the yield per unit area of land, which may have two effects on urban-rural income disparity. From the founding of the People's Republic of China to the 1990s, the shortage of agricultural products plagued China's socioeconomic development. Even after China's reform and opening up, commercial system was dominated by the state, and people had to use food stamps to buy food. When crop yields increased, rural residents could sell all kinds of grains in the nation's commercial system at relatively stable contract prices and even sell some grains at higher prices in free markets after they retained a sufficient amount of food for themselves. Under such circumstances, farmers' incomes increase with the increase in agricultural output, thereby narrowing urban-rural income disparity. After the 1990s, the continuously increasing agricultural production capacity solved food and clothing shortages. The supply of agricultural products was, to some extent, greater than the demand. As a result, the prices of agricultural products continued to decline after the mid-1990s, and the growth of farmers' incomes did not keep pace with the growth of agricultural products.

In addition to the prices of agricultural products, some factors directly related to the use of chemical fertilizers may also have resulted in an increase in yields without increasing farmers' incomes. Shen and Zhang (2007) found that the ratio between the total amount of chemical fertilizer application and the output value of the primary industry and the ratio between total power of agricultural machinery and the output value of the primary industry have significantly negative impacts on the disposable income per capita of rural households. The authors argued that the decline in the marginal productivity of chemical fertilizers and the increase in agricultural production costs were the main causes. In relation to these causes, the problem of the excessive application of chemical fertilizers has not been fully discovered. Based on random agricultural experiments in Kenya, Duflo, Kremer, and Robinson (2008) provided evidence that the application of chemical fertilizers at the correct doses yields high returns, whereas the return of non-optimal doses is negative. Hence, the return on chemical fertilizer application is sensitive to doses.

Although the overuse of chemical fertilizers is undesirable from the point of view of marginal productivity and costs, many factors, such as farmers' risk aversion, farmers' education level, and rural labor force migration ratio, make removing the dependency on the overuse of chemical fertilizers to increase farmers' incomes and narrow urban-rural income disparity difficult (Ebenstein, Zhang, McMillan, & Chen, 2011; Lamb, 2003; Zhou, Yang, Mosler, & Abbaspour, 2010). Therefore, the method of increasing farmers' incomes solely based on fertilizer application will not work in the long term.

Second, in the case of the tense population-land relationship in rural China, many rural surplus labor forces are released from their lands due to the productivity-enhancing effect of chemical fertilizer application (Lin, 1992; Wang & Xiao, 2008). Such an effect plays a key role in changing farmers' employment and income structures, causing farmers to seek career opportunities in cities and earn higher wage incomes. Although employment restrictions for migrant workers exist in the urban labor market (Sun, 2017), wage incomes have increasingly become a critical support for farmers escaping from poverty and achieving income growth (Cai & Du, 2011). According to the data released by the National Bureau of Statistics of China, at the national level, income of wages and salaries, agricultural income, net income from property, and net income from transfer in 2016 accounted for 40.6%, 38.3%, 2.2%, and 18.8% of the per capita disposable income of rural households, respectively. Compared with 2000, the proportion of rural residents' net business income decreased from 63.3% to 38.3% in 2016, while the proportion of rural residents' income of wages and salaries rose from 31.2% to 40.6% in 2016. Such a structural change in the per capita disposable income of rural households makes the correlation between rural households' disposable incomes and farmers' non-agricultural employment stronger. Farmers' non-agricultural employment closely relates to surplus labor forces' release effect caused by the application of chemical fertilizers. Sun (2017) argued that from 2001 to 2010, there was a convergence in the mechanism of wage determination between rural and urban labor forces in urban labor market. Sun (2017) also found a significant increase in the rate of return on education for migrant workers, and job segregation and wage discrimination for migrant workers were dramatically alleviated. Thus, the expected returns of rural surplus labor forces that migrate from rural areas to cities are continuously evolving in their favor.

Third, the application of chemical fertilizers provides grain and land supply guarantees for the development of urbanization and non-agricultural industries. On the one hand, the shift in economic structure from agricultural to non-agricultural industries requires an adequate grain supply and stable grain prices. The positive effect of chemical fertilizer application on yields provides a reliable basis for such grain security requirements. Zhang and Wan (2015) proved theoretically and empirically that the rate of surplus grain has a positive effect on urbanization. On the other hand, fertilizer application can increase the grain yields per hectare, sown area of crops, and multiple cropping index, which increases the degree of intensive land use and satisfies the demand of land use from the expansion of urban sectors and non-agricultural industries. Therefore, a theoretical inference is that the use of chemical fertilizers promotes the outflow of rural surplus labor by increasing agricultural productivity, the urbanization process, non-agricultural industry expansion, and development of the urban labor market through its effects on grain yields and intensive land use. The employment of rural surplus labor in cities can increase farmers' incomes and narrow urban-rural income disparity. However, the process of urbanization has had some negative impacts on rural land use, such as the loss of a large amount of arable land, the degradation of land quality, and the declining efficiency of land use, which are unfavorable for narrowing urban-rural income disparity.

Fourth, as the use of chemical fertilizers increases China's agricultural output and safeguards grain security, the government has gradually relaxed restrictions on farmers' cultivation decisions related to crops (Lin, 1992), mitigating farmers' dependency on grain production and helping them allocate resources, such as time, land, and water, more rationally. Farmers can then expand planting areas for cash crops, such as vegetables, fruits, and flowers; broaden the sources of rural household incomes; and increase their incomes.

Based on the aforementioned discussion, we expect chemical fertilizer application to narrow urban-rural income disparity. Such an effect has not been specifically tested in the literature. Additionally, the effect of chemical fertilizer application on rural-urban migration and providing guarantees for urbanization and the expansion of non-agricultural industries may theoretically be the most important for narrowing urban-rural income disparity. These two effects are also closely related to factors in the literature that affect the use of chemical fertilizers. Lamb (2003) found that farmers who increased labor supply for non-agricultural production in India effectively dampened agricultural production shocks and smoothed farmers' consumption. Additionally, the increase in the supply of non-agricultural labor leads to an increase in the intensity of chemical fertilizer application. Ebenstein et al. (2011) concluded that in China, the migration of rural labor to cities is crucial to explaining the excessive use of chemical fertilizers. Lin and Ma (2013) surveyed the wheat growers in the North China Plain and discovered that the extent of the overuse of chemical fertilizers per unit area by farmers was 116%. The marginal effect of fertilizer application on wheat yield is already low, which indicates the economic irrationality of farmers' use of chemical fertilizers. However, there must be a reasonable explanation for such a seemingly economic irrationality. After weighing the risk of agricultural production, cost of inputs, and expected return, farmers increase the intensity of chemical fertilizer application for risk aversion and income increases, which ultimately lead to farmers' dependence on chemical fertilizer application.

Finally, to increase income, smooth consumption, and avoid risks, farmers are motivated to use more chemical fertilizers. In other words, we observe that chemical fertilizer application is a necessary condition for farmers to increase their incomes. The increase in farmers' incomes, especially wage incomes, also depends on the non-agricultural employment opportunities of farmers and the degree of perfection of the labor market. In the process of China's urbanization, the migration of rural surplus labor to cities has been an irreversible trend. Compared with the non-agricultural labor market in rural areas, the degree of perfection of the urban labor market plays an increasingly important role in expanding non-agricultural employment and increasing farmers' wage incomes. Because urbanization and the development of the urban labor market are inseparable, and the expected higher urban wage income is the impetus for attracting rural surplus labor forces, the urbanization process is a sufficient condition for increasing farmers' incomes and

narrowing urban-rural income disparity. Therefore, in this empirical analysis, in addition to examining the effects of chemical fertilizer application and urbanization on urban-rural income disparity, we must also control for the impact of their interaction.

3. Methodology and data

Many macroeconomic variables, including urban-rural income disparity, are affected by their lagged levels and thus show a significant lag effect (Beck, Demirgüç-Kunt, & Levine, 2007; Roodman, 2009). To more precisely examine the direct impacts of chemical fertilizer application and urbanization and the impact of the interaction between these two variables on urban-rural income disparity, we establish the following regression equation:

$$gap_{it} = \beta_0 + \beta_1 l. gap_{it} + \beta_2 fer_{it} + \beta_3 cross_{it} + \beta_4 urban_{it} + \sum_j \alpha_j control_{jt} + u_i + y_t + e_{it} \quad (1)$$

where the subscripts i and t represent the i -th province and the t -th year, respectively; dependent variable gap_{it} is the ratio of disposable income per capita of urban and rural households, and $l.gap_{it}$ is its first-order lag; fer_{it} , $urban_{it}$, and $cross_{it}$ are the intensity of chemical fertilizer application, urbanization, and the intersection term, respectively; $control_{jt}$ is a set of the control variables; y_t is the time-specific effect; u_i is the province-specific effect; and e_{it} is the random error term.

One of the advantages of using the dynamic panel estimator is that it can control the endogeneity of all explanatory variables by using the instrument variables based on the lagged terms of explanatory variables. To test the robustness of the estimation results, we re-estimate the regressions by partitioning the sample into different regions and different years. Moreover, the relationship between the intensity of chemical fertilizer application and urban-rural income disparity may be driven by a reverse causation. For example, an expansion in urban-rural disparity may stimulate more rural-urban migration which in turn may stimulate the demand for chemical fertilizers. Although the dynamic panel estimator uses the “internal” instrument variables, i.e., the previous realizations of the explanatory variables, it may still suffer some problems, such as too many instruments (Roodman, 2009). It seems better to use an “external” instrument to check the robustness of estimation results. As Li (2014) pointed out, the amount of chemical fertilizers used in grain production accounts for over half of the total fertilizer application in China. Thus, as the dynamic panel estimator treats the previous realizations of the explanatory variables as instrument variables, we use the first-order lagged term of chemical fertilizer application per unit of grain as the instrument variable (IV) of the intensity of chemical fertilizer application. We also use the process of successively adding the control variables in baseline regressions to check the robustness of the core independent variables' coefficients and the possible disturbances caused by each control variable.

We use the system generalized method of moments (GMM), proposed by Blundell and Bond (1998), and a one-step method to estimate the coefficients of the dynamic panel data model. Regarding the dynamic panel data model, when the number of cross-sections is small, the differential estimator and system estimator are inefficient. Arellano and Bond (1991) and Blundell and Bond (1998) have indicated that although the two-step estimator is more effective, the asymptotic inference of the one-step standard error is more reliable. The problem of inefficiency may be more serious if the number of instrument variables is greater than or equal to the number of cross-sections. Due to data availability, Xizang (Tibet) Autonomous Region is excluded in our research sample. Thus, our panel consists of data for 30 provinces of China over the period 1997–2015. Hence, it is difficult to effectively avoid the problem that the number of instrument variables exceeds the number of cross-sections when we add more control variables into the regression equation. Based on the above reasons and following the suggestions proposed by Bond (2002), we mainly use the one-step method to estimate the coefficients of regression equations.

Table 1 summarizes the statistics of the variables. Based on the online database of the National Bureau of Statistics of China (NBS) and the statistical yearbooks of each province, we construct the dependent variable gap as the ratio between urban and rural households' disposable income per capita. In 2013, the NBS implemented a new urban-rural integrated household survey system; since 2014, the NBS has issued new income and expenditure data for urban and rural households. Such a statistical change may result in the data before and after 2013 being incomparable. To avoid measurement errors, we conduct a robustness check by splitting the sample into different years. Since many provinces adopted the NBS's new survey system when they released the data in 2013, we use a panel over the period 1997–2012 for the robustness check. Because China's accession to the World Trade Organization (WTO) in 2001 may have affected China's agricultural production, we also use a panel over the period 2002–2015 for the robustness check. In addition, it is desirable to replace the dependent variable gap by the Gini coefficient to measure urban-rural income disparity in robustness checks. To calculate the urban-rural Gini coefficient, we need collect a large number of related data. However, in the process of collecting and collating the related statistical data at China's provincial level, we encounter three tricky problems on the integrality and availability of related data. First, the statistical caliber changed during the sample period which is mentioned above. Second, even in the period of 2002–2012, when the statistical caliber was relatively stable, the problem of missing data is serious. Provincial-level regions with missing data during 2002–2012 include Tianjin, Jilin, Shandong, Hunan, Hainan, Xizang (Tibet), and Ningxia. Hebei, Guizhou, Gansu, and Qinghai did not offer complete data. Third, the income grouping criteria are different among provinces. For example, some provincial-level regions provide income data grouped by a five-equal-part method (such as Beijing and Shanghai), while others use a seven-equal-part method (such as Heilongjiang), or even use a non-equal method to group income data (such as Shanxi). Such problems lead to a certain degree of confusion in the calculation of the Gini coefficient. Hence, due to the limitation of the availability and integrality of data, we have to use variable gap to measure the urban-rural income disparity.

We are interested in the coefficients of the intensity of chemical fertilizer application (fer), urbanization ($urban$), and the intersection term ($cross$). According to the theoretical analysis, the intensity of chemical fertilizer application may narrow urban-rural income disparity, while urbanization may increase rural households' incomes through rural-urban migration and increase urban

Table 1
Summary statistics of variables.

Variable	Description	Mean	Std. Dev.
<i>gap</i>	Urban-rural income disparity = per capita disposable income of urban households/per capita disposable income of rural households	2.8491	0.6157
<i>fer</i>	Intensity of chemical fertilizer application = volume of the effective component of chemical fertilizers/total sown areas of farm crops, tons per hectare	0.3205	0.1117
<i>urban</i>	Urbanization = urban population/total population	0.4735	0.1577
<i>cross</i>	Intensity of chemical fertilizer application $\times$ Urbanization	0.1601	0.0944
<i>rgdp</i>	Real GDP per capita, 10,000 RMB, 1978 = 100	0.5993	0.5455
<i>inv</i>	Investment ratio = total investment in fixed assets/GDP	0.5350	0.2210
<i>st</i>	Industrial structure = added value of second industry/added value of tertiary industry	1.1550	0.3044
<i>open</i>	Trade dependency = total value of imports and exports/GDP	0.3103	0.3848
<i>fdi</i>	Foreign direct investment = FDI/GDP	0.0284	0.0273
<i>gov</i>	Government fiscal expenditure = local governments expenditure/GDP	0.1775	0.0867
<i>fd</i>	Financial development = total loans of fund uses of financial institutions/GDP	1.1073	0.3586
<i>edu</i>	Enrollment of senior secondary school students = number of total enrollment of students, senior secondary schools/total population, 100 students per 10 thousand population	1.5496	0.5236
<i>pest</i>	Pesticide intensity = use of agricultural pesticide/total sown areas of farm crops, tons per hectare	0.0106	0.0080
<i>film</i>	Plastic film intensity = use of agricultural plastic film/total sown areas of farm crops, 100 kg per hectare	0.1455	0.1204
<i>power</i>	Power of agricultural machinery = total power of agricultural machinery/total sown areas of farm crops, 100 kw per hectare	0.0494	0.0274
<i>irr</i>	Irrigation intensity = total irrigated area/total sown areas of farm crops, hectare per hectare	0.1909	0.1432
<i>ele</i>	Rural electricity consumption = electricity consumed in rural areas/total sown areas of farm crops, 100 million kwh per hectare	0.1683	0.2795

residents' incomes through scale economies in urban production and the wealth effect of real estate appreciation. Both [Lu and Chen \(2006\)](#) and [Chen and Lin \(2013\)](#) found that urbanization had a significant effect on reducing urban-rural income disparity, but [Zhong \(2011\)](#) did not observe such an effect. Notably, there is a certain degree of endogenous relationship between urban-rural income disparity and urbanization. For example, an increase in urban-rural disparity may stimulate more rural-urban migration which in turn may stimulate the process of urbanization. To alleviate such an endogeneity, besides using the dynamic panel estimator, we build another urbanization indicator by using the nighttime light data from satellite monitoring and use the first-order lagged term of such an urbanization indicator as the instrument variable for explanatory variable *urban* (see [Table 1](#)) which is calculated by using population data. The raw nighttime light data come from the rasterized satellite monitoring data released by the National Geophysical Data Center of the National Oceanic and Atmospheric Administration (NOAA) in the U.S., which reflect global nighttime light distribution and brightness, as well as economic activities at night. [Elvidge et al. \(1997\)](#), [Imhoff, Lawrence, Stutzer, and Elvidge \(1997\)](#), [Elvidge et al. \(1999\)](#), [Small, Elvidge, Balk, and Montgomery \(2011\)](#), and [Zhang and Seto \(2011\)](#) all believe that the nighttime light data can be used to measure the real level of urbanization. Currently, NOAA's global nighttime light data include four types, i.e., stable lights, average visible, average lights X Pct, and cloud free coverages. Following [Chen, Zhu, Shi, and Toshiaki \(2003\)](#), we select the most commonly-used stable lights data as basic data to measure the level of urbanization and construct a composite light index that can comprehensively reflect the breadth (average light intensity) and depth (regional light area) of urbanization based on data availability. Because data calibration deviation was caused by the satellite replacement before 1998 and the latest year of NOAA's nighttime light data is 2013, the year span of the light composite index is 1998–2013. Thus, we use a panel data set over the period 1998–2013 for the robustness check.

In addition, although chemical fertilizer application may increase labor productivity and release rural surplus labor forces, farmers' incomes cannot increase significantly without improvements in the urban labor market. In recent years, the institutional barriers of migration in China have weakened. The protection of the rights and interests of migrant workers in urban employment, health care, and social security have increased. The uncertainty of migrant workers' work and life in cities has declined significantly. Therefore, validating the interaction between chemical fertilizer application and the urbanization rate is essential.

We control for other variables that may affect urban-rural income disparity as follows.

(1) The real GDP per capita, which is used to control the impact of economic development on urban-rural income disparity. [Zhong \(2011\)](#) concluded that the real GDP per capita has a significant effect on expanding urban-rural income disparity. [Chen and Lin \(2013\)](#) found that real GDP per capita has a U-shaped effect on urban-rural income disparity, and the majority of the sample is in the stage where urban-rural income disparity declines as real GDP per capita increases.⁵ From the perspective of the practice of China's reform and opening up, with the development of the economy, urban-rural income disparity has shown a tendency of continuous expansion. However, with the increasing labor mobility, reforms of the income distribution system, and the continuous deepening of industries and cities nurturing agricultural and rural areas, the relationship between the level of economic development and urban-rural income disparity cannot be discussed only theoretically and empirical identification is necessary.

(2) Following [Zhong \(2011\)](#), we introduce the ratio of the added values between the second and tertiary industry to control for the income distribution effect of industrial structure changes. Urbanization is driven by the structural shift from agriculture to modern industrial and service sectors ([Davis & Henderson, 2003](#)), and such a shift may narrow urban-rural income disparity through

⁵ We attempted to introduce a square term of real GDP per capita in the regression model, but its effect was not statistically significant.

urbanization. The secondary and tertiary industries per se located in cities can absorb rural surplus labor forces and increase their income. However, due to the rural-urban migration restrictions imposed by the household registration system, *hukou*-based discrimination, and the urban-rural educational gap in China, the labor force demands in secondary and tertiary industries may be biased toward urban residents who usually possess a better educational background and working skills, even though the rural labor forces are much cheaper than their urban counterparts in urban labor market. For example, most rural labor forces cannot be employed by the state-owned enterprises in cities (Lu & Chen, 2006; Zhong, 2011). As a result, the development of secondary and tertiary industries may expand urban-rural income disparity. In addition, Chen and Lin (2013) asserted that the prioritized development strategy of the capital-intensive sectors widens urban-rural income disparity. Because most capital-intensive sectors belong to the secondary industry, controlling for the ratio of the added value between the secondary and tertiary industries can identify such effects to some extent.

(3) The degree of China's openness to the world is measured by trade dependence (*open*) and foreign direct investment (*fdi*). A common argument is that because most of the tradable goods are manufactured in cities, the increasing openness may widen urban-rural income disparity (Chen & Lin, 2013; Lu & Chen, 2006; Zhong, 2011). However, rural household incomes will increase as the export of agricultural products increases, and many rural labor forces are absorbed by urban manufacturing sectors. Therefore, the coefficients of trade dependence and foreign direct investment also need empirical identification.

(4) Physical capital (*inv*) and human capital (*edu*). The accumulation of physical capital enhances production capacity and absorbs a substantial amount of rural labor forces. Infrastructure, such as railways and highways, reduces transaction costs and facilitates large-scale labor migration across regions. However, because the physical capital investments are thought to be urban-biased in China, we expect that physical capital accumulation may expand urban-rural income disparity. In terms of human capital investment, some scholars (e.g., Zhong, 2011) have found that education can narrow urban-rural income disparity. We expect the coefficient of the human capital variable to be negative.

(5) The role of the government in the economy is measured by the share of government expenditures in GDP (*gov*). The role of the government is two-fold. First, the urban-biased fiscal expenditures may be detrimental to narrowing urban-rural income disparity. Second, while urban-rural income disparity increases, local governments should use fiscal instruments to moderate this gap.

(6) Financial development (*fd*). Chen et al. (2010) found evidence that financial development is not conducive to narrowing urban-rural income disparity, but other studies have indicated that the impact of financial development on urban-rural income disparity is not significant (Lu & Chen, 2006).

(7) Agricultural modernization, such as building agricultural irrigation system, increasing agricultural mechanization input, and using mulch and vegetable greenhouses, can improve agricultural productivity. However, the reform of market-oriented system in rural areas still faces a lot of difficulties, and the overall improvement of agricultural production is slow (Zhu, 2011). Especially, the phenomenon of rural poverty in some regions in China is still prevalent (Lohmar, Gale, Tuan, & Hansen, 2009). Meanwhile, with a lower rural income level and less arable land per capita, China's agricultural production still retains a traditional mode, to a certain extent, and small-scale rural household production patterns are subject to more resource constraints (Zhu, 2011). Such factors limit the use of modern agricultural production inputs. In addition, the statistical data regarding the mulch and vegetable greenhouses, high-yield seeds, and fiscal subsidies for agricultural production technology are not released at the provincial level or suffer a serious missing problem. Thus, we control the intensity of pesticide application (*pest*), the intensity of agricultural plastic film application (*film*), agricultural machinery power (*power*), effective irrigation areas (*irr*), and rural electricity consumption (*ele*) as additional control variables. To some extent, agricultural machinery power and rural electricity consumption can be used to measure the agricultural mechanization inputs, effective irrigation areas can be used to measure the development of agricultural irrigation system, and agricultural plastic film application can be used to measure the status of mulch and vegetable greenhouses. These variables represent the improvements in agricultural production and/or rural living conditions, which may improve the efficiency of agricultural production, increase farmers' incomes, and narrow urban-rural income disparity.

Liu and Huang (2013) argued that Chinese farmers use excessive amounts of pesticides to avoid agricultural production risks, but this practice helps ensure the stability of agricultural output and income. Investment in agricultural machinery is an example of technological progress in agricultural production, which saves the agricultural labor force, promotes rural-urban migration, and increases farmers' incomes (Van den Berg, 2007). As another key factor that causes serious agricultural pollution, the application of agricultural plastic film has an inverted U-shaped relationship with agricultural income per capita, which is positively related to income inequality in rural areas (Du & Luo, 2013). Bhattacharai, Sakthivadivel, and Hussain (2002) found that the use of irrigation systems can effectively reduce rural poverty; however, increasing agricultural output by irrigation systems is a necessary but insufficient condition for alleviating rural poverty, and its impact on poverty alleviation depends on the development of the rural labor market, wage structure, and other factors. Finally, rural electricity consumption may have a significant impact on agricultural and non-agricultural production activities in rural areas and is therefore a potential variable that affects farmers' income (Fan & Zhang, 2004).

4. Empirical analysis

4.1. Benchmark regression results

To alleviate the endogeneity problem, we use the system GMM panel data estimator. By adding the control variables successively, Regression (2.9) in Table 2 eventually includes all the explanatory variables. Notably, although the GMM methodology can derive a consistent estimator, the GMM estimator is prone to bias due to sample size, weak instrument variables, and other issues. For this

Table 2
Benchmark regression results.

Independent variable	(2.1)	(2.2)	(2.3)	(2.4)	(2.5)	(2.6)	(2.7)	(2.8)	(2.9)
OLS	0.9458	0.9485	0.9498	0.9485	0.9472	0.9430	0.9433	0.9438	0.9281
FE	0.8422	0.8341	0.8356	0.8356	0.8127	0.8109	0.8095	0.8129	0.7963
<i>Lgap</i>	0.8878*** (0.1491)	0.9257*** (0.0839)	0.9139*** (0.0613)	0.8959*** (0.0548)	0.8985*** (0.0539)	0.9256*** (0.0475)	0.9260*** (0.0476)	0.9366*** (0.0488)	0.9249*** (0.0987)
<i>fer</i>	−2.8274* (1.5690)	−1.6581*** (0.6216)	−1.5908*** (0.5422)	−1.7271*** (0.5560)	−1.7795*** (0.6763)	−1.5733*** (0.6717)	−1.5744*** (0.6747)	−1.3779*** (0.6439)	−2.2115* (1.1512)
<i>urban</i>	−2.3368* (1.2096)	−1.5493*** (0.5544)	−1.4796*** (0.4962)	−1.5999*** (0.4980)	−1.6610*** (0.6391)	−1.4858*** (0.6355)	−1.4832*** (0.6312)	−1.2832*** (0.6168)	−1.9861* (1.2022)
<i>cross</i>	5.5187** (2.6593)	3.2371*** (1.0186)	3.0557*** (1.0394)	3.2892*** (1.1336)	3.3537** (1.3371)	3.0910** (1.2758)	3.0944** (1.2875)	2.7376** (1.1722)	4.1582** (1.9509)
<i>rgdp</i>		0.0710*** (0.0262)	0.0700*** (0.0261)	0.0703** (0.0282)	0.0682** (0.0279)	0.0745*** (0.0280)	0.0746*** (0.0280)	0.0662* (0.0376)	0.0867 (0.0658)
<i>inv</i>			−0.0157 (0.0894)	−0.0064 (0.0898)	−0.0105 (0.1155)	−0.0450 (0.1355)	−0.0441 (0.1357)	−0.0826 (0.1119)	−0.0083 (0.0864)
<i>st</i>				0.0040 (0.0334)	0.0093 (0.0282)	0.0277 (0.0263)	0.0268 (0.0267)	0.0301 (0.0292)	0.0549** (0.0259)
<i>open</i>					0.0188 (0.0499)	0.0034 (0.0591)	0.0040 (0.0571)	−0.0083 (0.0533)	0.0171 (0.0686)
<i>fdi</i>					0.1365 (0.3972)	0.1959 (0.3905)	0.1882 (0.3848)	0.1439 (0.3748)	0.7487 (0.7140)
<i>gov</i>						0.1280 (0.1480)	0.1313 (0.1818)	0.1532 (0.1477)	0.0117 (0.1579)
<i>fd</i>							−0.0026 (0.0342)	−0.0024 (0.0316)	0.0242 (0.0548)
<i>edu</i>								−0.0015 (0.0346)	0.0353 (0.0540)
<i>pest</i>									1.4950* (0.8957)
<i>film</i>								0.0404 (0.1177)	0.0404 (0.1177)
<i>power</i>								−0.6835 (0.5766)	−0.6835 (0.5766)
<i>irr</i>								0.0271 (0.1573)	0.0271 (0.1573)
<i>ele</i>								−0.0088 (0.0369)	−0.0088 (0.0369)
Constant	1.3234 (0.9678)	0.8413* (0.4755)	0.9191** (0.3731)	0.9445*** (0.3213)	1.0497*** (0.3709)	0.7711** (0.3534)	0.7723** (0.3581)	0.6694* (0.3676)	0.9905 (0.8135)
AR (1)	−3.10 [0.002]	−3.61 [0.000]	−3.93 [0.000]	−3.91 [0.000]	−3.98 [0.000]	−4.15 [0.000]	−4.15 [0.000]	−4.14 [0.000]	−3.64 [0.000]
AR (2)	0.29 [0.772]	0.23 [0.818]	0.23 [0.816]	0.25 [0.801]	0.26 [0.791]	0.22 [0.825]	0.22 [0.826]	0.19 [0.848]	0.26 [0.793]
Sargan test	1.40 [0.496]	1.09 [0.581]	1.21 [0.751]	1.32 [0.723]	1.46 [0.692]	1.37 [0.714]	1.35 [0.716]	1.84 [0.765]	3.08 [0.215]

Notes: (1) Dependent variable: urban-rural income disparity; (2) the numbers in parentheses are robust standard errors; the numbers in square brackets are *p*-values; (3) ***, **, and * mean significance at the levels of 1%, 5%, and 10%, respectively; (4) OLS and FE represent the estimated coefficients of *Lgap* by using the ordinary least-squares and the fixed-effect model, respectively; (5) observations are 540 and groups are 30; (6) all regressions control for the individual-specific effect and time-specific effect.

reason, we use the rule of thumb suggested by Bond (2002) to assess the presence of a large bias in the GMM estimator. Bond (2002) argued that the coefficients of the first-order lagged dependent variable estimated by the dynamic FE and OLS have downward and upward biases, respectively, and the coefficient of the first-order lagged dependent variable estimated by the system GMM should be greater than FE and smaller than the OLS estimation. In Table 2, we observe that the coefficients of *l.gap* in all regressions are greater than those obtained by using the dynamic FE and smaller than those obtained by using the OLS. This result indicates that the system GMM methodology does not produce significantly biased estimates. Furthermore, the results of the Sargan test and the second order serial correlation test in Table 2 indicate that our estimation is reasonable.

The set of regressions presented in Table 2 offer evidence that the intensity of chemical fertilizer application (*fer*) and urbanization (*urban*) have significant effects on narrowing urban-rural income disparity; the former is consistent with the theoretical hypothesis of this paper, and the latter is consistent with much of the literature (Chen & Lin, 2013; Lu & Chen, 2006). Based on Regression (2.9), when the interaction term (*cross*) is not taken into account, the urban-rural income disparity will decrease by 0.22, with each standard deviation increase in the intensity of chemical fertilizer application. Similarly, the urban-rural income disparity decreases by 0.32, with urbanization increasing by one standard deviation. These results imply that, on the one hand, given that the level of urbanization in China remains low, there is room to narrow urban-rural income disparity by promoting urbanization, and on the other hand, although increasing the intensity of chemical fertilizer application mitigates urban-rural income disparity, continuing to increase the intensity is not preferred because China's intensity of chemical fertilizer application is already far above the world average; notably, the overuse of chemical fertilizers has caused serious environmental pollution and hazards to human health in China. To address such a dilemma, we seek the answer from the coefficient of the interaction term.

The results in Table 2 also indicate that the coefficients of the interaction term (*cross*) are significantly positive. Hence, combining the non-interacted term (*fer*) and interacted term (*cross*) allows us to identify the threshold of urbanization whereby, below (above) the threshold, increasing (decreasing) the intensity of chemical fertilizer application will mitigate urban-rural income disparity. Based on Regression (2.9), when the urbanization level is taken as the sample average (0.4735), after considering the interaction item, the urban-rural income disparity only decreases by 0.024, with the intensity of chemical fertilizer application increasing by one standard deviation. The intensity of chemical fertilizer application now has little effect on narrowing urban-rural income disparity. We can also calculate a threshold value of urbanization which is 0.5318 based on Regression (2.9). Below (above) 0.5318, the increase (decrease) in the intensity of fertilizer application has the effect on narrowing urban-rural income disparity.

The main result of benchmark regressions is that the intensity of the chemical fertilizer application has a significant impact on urban-rural income disparity, but the effect depends critically on the level of urbanization. This result provides a novel interpretation for the impacts of inputs of agricultural production and urbanization on urban-rural income disparity and has valuable policy implications. We argue that as the process of urbanization advances, wage income has become the main source of rural household income (Cai & Du, 2011). Since rural households are increasingly dependent on employment in urban sectors rather than using additional chemical fertilizers in agricultural production to achieve income growth, the process of urbanization is gradually changing the combination of inputs of agricultural production through an income growth mechanism. For farmers, the method of increasing their incomes by simply relying on increasing the inputs of agricultural factors is replaced by the income-increasing opportunities from urbanization. In view of the obvious damages caused by agrochemicals to environmental quality, agro-ecology, food safety, grain security, and human health, the continuous progress of urbanization has resulted in two welfare improvement effects: the effect of urbanization per se on narrowing urban-rural income disparity, and the effect of urbanization on alleviating chemical toxicities by affecting the inputs of agrochemicals. Therefore, steadily advancing the process of urbanization has important practical significance for increasing farmers' incomes, alleviating the unfair income distribution between urban and rural areas, and providing a means to reduce the dependence of agricultural production on chemical fertilizers and other chemical inputs and mitigate the pollution and health damages caused by agrochemicals. Moreover, in this sense, the two zero-growth action plans issued by the MOA have strong rationality and feasibility under the conditions of continuous urbanization.

For the control variables, the increase in GDP per capita widens urban-rural income disparity. This result implies that in the course of China's economic development, the growth rate of urban and rural residents' income has been unequal. The use of pesticides has widened urban-rural income disparity. This feature is of great significance regarding the overuse of pesticides in China. Thus, the suppression of the excessive use of pesticides narrows urban-rural income disparity and benefits the environment and health. The increase in the ratio of the value added of the secondary industry to the value added of the tertiary industry also widens urban-rural income disparity. This result can be attributed to the following: as the secondary industry's skill requirements for workers increases, the expansion of the secondary industry adversely affects the employment of low-skilled rural surplus labor forces.

4.2. Robustness checks

First, according to the classification of the National Bureau of Statistics of China, we divide our sample into three regions: eastern, middle, and western regions of China. China's main agricultural production areas are located in the southeast of the 15-in. isohyet. Most agricultural production in western provinces, such as Qinghai, Xinjiang, Ningxia, and Inner Mongolia, are based on animal husbandry. The averages of the intensity of fertilizer application in eastern (407 kg/ha), middle (295 kg/ha) and western regions (253 kg/ha) of China decreased in turn. On the whole, the intensity of chemical fertilizer application in western region is relatively low, and natural environment in western region is not very suitable for crop growth. Therefore, we expect that the impacts of the intensity of chemical fertilizer application in eastern, middle and western regions on urban-rural income disparity may be different.

Regression (2.9) in Table 2 is re-estimated regarding these three regions. The results of Regressions (3.1)–(3.3) in Table 3 are discussed as follows. First, because most of China's grain producing areas are located in eastern and middle regions, while in western

Table 3
Robustness check results (different regions and different years).

Independent variable	(3.1)	(3.2)	(3.3)	(3.4)	(3.5)	(3.6)
	Eastern	Middle	Western	Time effect	1997–2012	2002–2015
OLS	0.9029	0.7215	0.8792	0.9151	0.9403	0.9134
FE	0.6961	0.5191	0.7310	0.8915	0.7604	0.6986
<i>Lgap</i>	0.8678*** (0.0840)	0.6937*** (0.0750)	0.7383*** (0.1190)	0.8990*** (0.0982)	0.8587*** (0.0771)	0.9048*** (0.0487)
<i>fer</i>	−3.6819** (1.4580)	−2.7047* (1.5538)	1.9741 (1.7035)	−4.6084*** (1.6093)	−1.9062*** (0.7342)	−1.2559** (0.5455)
<i>urban</i>	−2.4984* (1.3357)	−1.7193** (0.8410)	0.3749 (1.1274)	−3.9119** (1.5452)	−1.6303** (0.6406)	−1.4263*** (0.5089)
<i>cross</i>	5.5731*** (2.1580)	6.5208** (3.2670)	−3.4316 (3.9463)	8.1813*** (2.8561)	4.0250*** (1.4444)	2.4940*** (0.9118)
<i>rgdp</i>	0.0015 (0.0556)	−0.6586** (0.2748)	0.2653 (0.1962)	0.1418* (0.0822)	−0.0458 (0.0765)	0.0229 (0.0415)
<i>inv</i>	−0.0349 (0.1437)	−0.3587** (0.1462)	−0.2873 (0.2844)	−0.0730 (0.1473)	0.0286 (0.0886)	−0.0933 (0.0725)
<i>st</i>	−0.1190 (0.1142)	0.1384** (0.0618)	−0.1226 (0.1490)	0.0723 (0.0609)	0.0522 (0.0377)	0.0099 (0.0286)
<i>open</i>	0.0566 (0.1148)	−0.1365 (0.4296)	−1.3495*** (0.3948)	0.1036 (0.0984)	−0.1586 (0.0989)	−0.0474 (0.0638)
<i>fdi</i>	1.2081** (0.6073)	−4.1679*** (1.2771)	1.2523 (3.0044)	2.0067** (0.9554)	0.9242 (0.7831)	1.9393** (0.8474)
<i>gov</i>	−0.0820 (0.3077)	1.3609 (1.1448)	0.4776 (0.7022)	−0.1480 (0.2381)	0.1874 (0.2630)	0.2040 (0.1432)
<i>fd</i>	−0.2378* (0.1395)	0.0119 (0.0410)	−0.0560 (0.1700)	0.0239 (0.0783)	0.0294 (0.0619)	0.0036 (0.0680)
<i>edu</i>	−0.0588 (0.0799)	0.0860 (0.0718)	−0.1328** (0.0639)	0.1242 (0.0867)	−0.0706 (0.0565)	−0.0590*** (0.0207)
<i>pest</i>	5.8703* (3.1804)	9.0782** (4.6277)	−4.6076 (6.2137)	4.7111 (3.4888)	0.6928 (1.6759)	−0.9885 (0.8600)
<i>film</i>	0.5099 (0.3969)	0.4098 (1.6515)	0.9379** (0.4738)	0.0078 (0.2149)	0.2267 (0.1649)	0.1411 (0.1116)
<i>power</i>	0.0895 (0.8366)	−0.0261 (1.0128)	−1.6251 (2.1439)	−0.9854 (0.8463)	−1.0643** (0.5077)	−0.5284 (0.5151)
<i>irr</i>	−0.0406 (0.2702)	0.3467 (0.2489)	−1.0090* (0.6105)	0.1135 (0.1889)	−0.1265 (0.1438)	−0.0275 (0.1237)
<i>ele</i>	0.0760 (0.0800)	−0.5641 (0.4561)	0.6276 (0.6877)	−0.0732 (0.0794)	0.1081* (0.0601)	0.0347 (0.0344)
Constant	1.9881** (0.9804)	1.1732*** (0.4358)	1.1001 (0.7300)	1.8969** (0.8142)	1.1033** (0.4706)	1.0387** (0.4141)
Obs. (group)	198 (11)	144 (8)	198 (11)	540 (30)	450 (30)	420 (30)
AR (1)	−2.22 [0.027]	−2.27 [0.023]	−2.19 [0.029]	−3.57 [0.000]	−4.02 [0.000]	−3.81 [0.000]
AR (2)	−1.51 [0.131]	0.91 [0.362]	0.03 [0.974]	−1.50 [0.134]	−0.12 [0.901]	0.17 [0.866]
Sargan test	2.16 [0.540]	9.67 [0.139]	1.31 [0.971]	2.30 [0.513]	6.59 [0.253]	9.27 [0.320]

Notes: (1) Dependent variable: urban-rural income disparity; (2) the numbers in parentheses are robust standard errors; the numbers in square brackets are p-values; (3) ***, **, and * mean significance at the levels of 1%, 5%, and 10%, respectively; (4) OLS and FE represent the estimated coefficients of *Lgap* by using the ordinary least-squares and the fixed-effect model, respectively; (5) Regression (3.4) does not control for the time-specific effect, and other regressions control for both the individual-specific effect and time-specific effect.

region, the arable land is scarce and the climatic conditions are not conducive to the growth of crops, the increase in the intensity of chemical fertilizer application has a statistically significant effect on narrowing the urban-rural income disparity in eastern and middle regions but not in western region.

Second, the role of urbanization in narrowing urban-rural income disparity is significant in eastern and middle regions but not in western region. Because urbanization level in western region is relatively low (average values of urbanization in eastern, middle and western regions are 0.8231, 0.7641, and 0.3772, respectively) and the rural labor forces of western region usually migrate to eastern and middle regions, the urbanization process in western region may not increase local farmers' income growth. Third, in eastern and middle regions, but not in western region, the estimated coefficients of interaction terms are statistically significant. This is consistent with the aforementioned first and second findings.

Fourth, the intensity of pesticide application has a significant effect on widening the urban-rural income disparity in eastern and middle regions. The pesticide application in eastern and middle regions is much higher than in western region (average values of pesticide application in eastern, middle and western regions are 0.0167, 0.0100, and 0.0048 tons per ha, respectively). The overuse of pesticides has become an economic burden for rural households. Although effective irrigation areas can significantly narrow the urban-rural income disparity in western region, such an effect is not observed in middle region. On the one hand, since the natural

endowments for agricultural production in western region are relatively poor (Zhu, 2011), a high efficiency of water resource and arable land management can improve significantly with the expansion of effective irrigation areas, promoting crop growth. Such a scenario is conducive to raising rural households' incomes. On the other hand, Bhattarai et al. (2002) argued that the role of irrigation systems in rural poverty reduction depends on the development of labor market and the opportunities for non-agricultural employment. The provinces in middle region are the main sources of the outflow of rural labor forces to eastern region (Sun, 2017). This indicates that the local labor market in middle region fails to provide farmers with effective non-agricultural employment opportunities, thus limiting the role of irrigation system in increasing farmers' incomes. In addition, although urbanization rate is relatively low and local labor market may not provide sufficient non-agricultural employment opportunities in western region, the positive effect of the expansion of effective irrigation areas on rural households' incomes is dominated in western region.

For other control variables in Regressions (3.1)–(3.3), GDP per capita has a significant effect on narrowing the urban-rural income disparity in middle region. The openness in western region and the FDI in middle region play significant roles in narrowing urban-rural income disparity. In eastern region, FDI has a significant effect on widening urban-rural income disparity. This can be attributed to the FDI in eastern region, which prefers to capital-intensive industries and has higher skill requirements. Human capital plays a significant role in narrowing the urban-rural income disparity in western region. In middle region, increasing investment and absorbing more FDI are conducive to narrowing urban-rural income disparity.

As the second robustness test, we check whether the results are robust to the time-specific effect and different time periods. Regression (3.4) does not control for the time-specific effect that may affect all provinces in any year. Our main results hold. In 2013, the NBS implemented a new urban-rural integrated household survey system. To eliminate measurement errors caused by statistical changes, Regression (3.5) uses the 1997–2012 period for estimation. Regression (3.6) uses the 2002–2015 period to examine the impact of China's accession to the WTO in 2001 regarding the import and export of agricultural products and agricultural production, which likely affected the use of chemical fertilizers. Again, in both cases, our main results hold.

As the third robustness test, we use the IV method to deal with the endogeneity problem. Because the amount of chemical fertilizers used in grain production accounts for over half of the total fertilizer application (Li, 2014), Regression (4.1) in Table 4 uses the first-order lag of the intensity of chemical fertilizer application per unit of grain yield, which is measured as the amount of chemical fertilizer application divided by the total grain output as an instrumental variable for the intensity of chemical fertilizer application. Our results still hold. Regressions (4.2)–(4.4) re-estimate Regressions (3.1)–(3.3) by using the aforementioned instrumental variable. The estimation yields similar results.

Next, we construct another urbanization indicator by using the nighttime light data from satellite monitoring and use the first-order lagged term of the new urbanization variable as the instrument variable of explanatory variable *urban* which is calculated by using population data. In Table 4, Regressions (4.5) and (4.6)–(4.8) re-estimate Regressions (2.9) and (3.1)–(3.3) by using the IV method. The estimation still yields similar results.

All the results show that the previous benchmark regression results are robust; thus, the conclusions obtained are credible.

5. Mechanism analysis

The effect of chemical fertilizer application on urban-rural income disparity is dependent on urbanization, which implies there is a threshold of urbanization to be identified. Following Hansen (1999), we estimate the following panel threshold regression model:

$$g_{pit} = \varphi_0 + \varphi_1 urban_{it} + \varphi_{21} fer_{it} I(urban_{it} \leq \lambda) + \varphi_{22} fer_{it} I(urban_{it} > \lambda) + \sum_j \alpha_j \times control_{jt} + u_i + e_{it} \quad (2)$$

where φ_0 , φ_1 , φ_{21} , φ_{22} , and α_j are coefficients to be estimated; $urban_{it}$ is the threshold variable; λ is the single threshold value; $I(\cdot)$ is the indicator function; u_i is the individual-specific effect; and e_{it} is the random error term.

First, we attempted to identify the threshold effects based on the panel of 30 provinces over the period 1997–2015, but we did not obtain statistically significant results with one or more stable threshold values. Theoretically, the panel threshold effect works for large-N and small-T panel data. For example, Hansen (1999) used a panel of 565 companies over 15 years. The panel data used in this paper has relatively long time span and relatively small cross-section of individuals. Notably, both urbanization level and the intensity of chemical fertilizer application have been increasing rapidly since the 1990s. Moreover, agricultural production is sensitive to the changes in weather conditions, causing that the application of chemical fertilizers is also affected by unpredictable weather factors. Therefore, the threshold effect of urbanization may be prone to fluctuation over time.

To solve the problem that the threshold effect may not be easily captured over time, we first drop the observation of year 1997 from the original sample and construct a panel data set of 30 provinces over the period 1998–2015. Next, we transform our time series data into non-overlapping 3-year and 6-year averages, respectively. By using these two new data sets, we perform the panel threshold effect based on Eq. (2). Simultaneously, to alleviate the endogenous problem, we use the first-order lag values for all the independent variables in Eq. (2).

As shown in Table 5, a single threshold level of urbanization is 0.3734, with the F test value being statistically significant at the 10% level. Below (above) this threshold level, the effect of chemical fertilizer application on narrowing urban-rural income disparity is (is not) statistically significant. Another observation is that the estimated threshold level is not statistically significant at 5% level, and the coefficients of other control variables are not consistent with the estimation results above. This is most likely to occur because the urbanization indicator constructed by using population data in this study cannot be considered as a strictly exogenous variable.

Table 4
Robustness check results (IV method).

Independent variable	(4.1)	(4.2)	(4.3)	(4.4)	(4.5)	(4.6)	(4.7)	(4.8)
	<i>fer</i> IV all regions	<i>fer</i> IV Eastern	<i>fer</i> IV Middle	<i>fer</i> IV Western	<i>urban</i> IV all regions	<i>urban</i> IV Eastern	<i>urban</i> IV Middle	<i>urban</i> IV Western
OLS	0.9281	0.9029	0.7215	0.8792	0.9332	0.8838	0.6824	0.9022
FE	0.7963	0.6961	0.5191	0.7310	0.7736	0.6189	0.4848	0.7384
<i>Lgap</i>	0.9234*** (0.0835)	0.8571*** (0.0746)	0.7055*** (0.0685)	0.7632*** (0.1093)	0.8462*** (0.1185)	0.8303*** (0.1112)	0.6769*** (0.0850)	0.8559*** (0.0565)
<i>fer</i>	−1.7485** (0.8759)	−3.5689*** (1.3197)	−2.6149* (1.4495)	1.9843 (2.9939)	−3.1305** (1.3680)	−3.0938*** (0.5645)	−5.9538*** (1.6427)	2.9645 (2.1875)
<i>urban</i>	−1.5489* (0.8959)	−2.2873** (1.1658)	−1.6875** (0.7793)	0.5204 (1.1838)	−3.9880** (1.6265)	−3.0882*** (0.9998)	−2.9322*** (0.8176)	1.2374 (1.9296)
<i>cross</i>	3.2031** (1.5036)	5.1649*** (1.9291)	7.0123* (4.1799)	−3.1637 (6.8024)	5.2199** (2.0588)	4.1835*** (1.0137)	14.6750*** (3.3815)	−7.9477 (5.5129)
<i>rgdp</i>	0.0502 (0.0522)	−0.0223 (0.0392)	−0.8097* (0.4574)	0.2612 (0.2784)	0.2037* (0.1237)	0.2166** (0.1030)	−1.0269*** (0.3376)	−0.2606 (0.1866)
<i>inv</i>	0.0225 (0.0832)	−0.0219 (0.1277)	−0.3185** (0.1597)	−0.3161 (0.3526)	0.1117 (0.2159)	−0.0655 (0.3023)	−0.5478*** (0.2039)	0.1894 (0.1816)
<i>st</i>	0.0516** (0.0203)	−0.1184 (0.0982)	0.1500 (0.1059)	−0.1350 (0.1379)	−0.0595 (0.1214)	−0.2369** (0.1102)	0.2076*** (0.0597)	0.1763 (0.1095)
<i>open</i>	0.0128 (0.0494)	0.0764 (0.0741)	−0.2647 (0.3903)	−1.3195*** (0.2647)	0.3374* (0.1966)	0.1748 (0.1584)	0.3017 (0.9139)	−0.9671*** (0.2970)
<i>fdi</i>	0.4119 (0.5344)	1.0783* (0.6354)	−4.7408*** (1.4716)	1.4775 (2.4953)	1.5325 (1.4482)	1.8177* (1.0235)	−3.8091** (1.8005)	0.1923 (1.4262)
<i>gov</i>	0.0098 (0.1247)	−0.2291 (0.3772)	1.7388 (1.4727)	0.6940 (0.9426)	−0.8926 (0.6379)	−0.3119 (0.8298)	0.7313 (2.3762)	−1.1033*** (0.3319)
<i>fd</i>	0.0089 (0.0480)	−0.2515** (0.1155)	−0.0169 (0.0709)	−0.0594 (0.1667)	0.0702 (0.1203)	−0.1343 (0.1600)	−0.0707 (0.1186)	−0.0521 (0.1748)
<i>edu</i>	−0.0209 (0.0380)	−0.0970 (0.0842)	0.0945 (0.0789)	−0.1244** (0.0521)	0.2239 (0.1389)	0.2237 (0.1683)	0.1735* (0.0986)	−0.0556 (0.0824)
<i>pest</i>	1.4937** (0.6981)	6.9152** (2.9100)	7.9986** (4.0795)	−3.1958 (10.8630)	0.1122 (2.3625)	3.7042 (3.4177)	9.1572*** (3.4477)	−10.9959 (8.1679)
<i>film</i>	0.0843 (0.0988)	0.5025 (0.3388)	−1.0133 (1.8795)	0.8825*** (0.3310)	0.0644 (0.2772)	0.3751 (0.5309)	−1.1199 (0.9784)	0.6607 (0.5505)
<i>power</i>	−0.4707 (0.4415)	0.2410 (0.7590)	−0.0342 (0.8320)	−2.1323 (3.3735)	−0.3170 (0.8821)	0.5240 (0.6624)	4.3524 (2.9233)	0.2971 (1.7259)
<i>irr</i>	−0.0230 (0.1349)	−0.0395 (0.2142)	0.3118 (0.4119)	−0.9536 (0.5876)	−0.2436 (0.2842)	−0.1197 (0.3433)	1.0234*** (0.2195)	−0.5721 (0.4717)
<i>ele</i>	0.0086 (0.0316)	0.0702 (0.0877)	−0.6657 (0.7078)	0.6490 (0.7786)	−0.1215 (0.0837)	0.0053 (0.1013)	−1.7619** (0.7775)	−0.0414 (1.2433)
Constant	0.9097 (0.6587)	1.9812** (0.8622)	1.1859*** (0.3741)	0.8778 (0.6413)	1.9139** (0.9270)	1.9454*** (0.7293)	1.8460*** (0.5356)	0.0860 (0.6138)
Obs. (group)	540 (30)	198 (11)	144 (8)	198 (11)	480 (30)	176 (11)	128 (8)	176 (11)
AR (1)	−3.79 [0.000]	−2.34 [0.019]	−2.27 [0.023]	−2.54 [0.011]	−2.69 [0.007]	−2.11 [0.035]	−2.04 [0.041]	−2.53 [0.012]
AR (2)	0.25 [0.803]	−1.45 [0.148]	1.05 [0.295]	−0.03 [0.979]	0.24 [0.813]	−1.58 [0.114]	0.41 [0.683]	−0.71 [0.480]
Sargan test	3.91 [0.272]	4.03 [0.259]	8.41 [0.209]	1.47 [0.961]	5.04 [0.411]	3.10 [0.797]	8.44 [0.207]	3.84 [0.698]

Notes: (1) Dependent variable: urban-rural income disparity; (2) the numbers in parentheses are robust standard errors; the numbers in square brackets are p-values; (3) ***, **, and * mean significance at the levels of 1%, 5%, and 10%, respectively; (4) OLS and FE represent the estimated coefficients of *Lgap* by using the ordinary least-squares and the fixed-effect model, respectively; (5) All regressions control for both the individual-specific effect and time-specific effect; (6) The sample period of Regressions (4.1)–(4.4) is 1997–2015 and the sample period of Regressions (4.5)–(4.8) is 1998–2013.

For example, the expansion of urban-rural income disparity may stimulate more rural-urban migration to accelerate urbanization process. Using the dynamic panel estimator and GMM, such an endogeneity problem can be alleviated to some extent. The benchmark regression results in Table 2 indicate that the effect of chemical fertilizer application on narrowing urban-rural income disparity depends on urbanization level. To examine such a threshold effect, we use the panel threshold regression model (Hansen, 1999). However, Hansen (1999) suggested that an exogenous variable should be used. Hence, we adopt the first-order lagged terms for all independent variables, including urbanization, to meet such a requirement. Because Hansen's (1999) panel threshold regression model is a static panel regression model, and we cannot effectively control the endogeneity problem in the model, we use the first-order lagged terms for all independent variables. Although the threshold regression results may be biased and different from those of the dynamic panel data model, we have found some evidence that urbanization is a threshold variable, which is consistent with the findings of the benchmark regression results.

Table 5

Estimation results of the threshold effect considering non-overlapping 3-year averages.

Independent variable	Coefficient	Independent variable	Coefficient	Independent variable	Coefficient	Independent variable	Coefficient
<i>Lurban</i>	−2.6667** (1.0438)	<i>Lfdi</i>	−2.0531 (2.2070)	<i>Lfilm</i>	1.0554* (0.6173)	<i>Lfer*I (Lurban ≤ 0.3734)</i>	−2.0253* (1.0419)
<i>Lrgdp</i>	0.2846* (0.1437)	<i>Lgov</i>	−3.0067*** (0.8180)	<i>Lpower</i>	1.8841 (2.2772)	<i>Lfer*I (Lurban > 0.3734)</i>	−1.5706 (0.9674)
<i>Linv</i>	−0.1928 (0.3058)	<i>Lfd</i>	−0.1491 (0.1639)	<i>Lirr</i>	−2.0592* (1.1724)	Threshold value	0.3734
<i>Lst</i>	−0.2248 (0.1636)	<i>Ledu</i>	0.3493*** (0.0715)	<i>Lele</i>	−0.0573 (0.2209)	F test [p-value]	6.39 [0.0733]
<i>Lopen</i>	0.1220 (0.2057)	<i>Lpest</i>	4.4731 (5.7054)	Constant	5.1572*** (0.4947)	Obs. (group)	150 (30)
						R ² (within)	0.5977

Notes: (1) Dependent variable: urban-rural income disparity; (2) the numbers in parentheses are robust standard errors; the numbers in square brackets are *p*-values; (3) ***, **, and * mean significant at 1%, 5%, and 10%, respectively.

In Table 6, there is a statistically significant single threshold level of 0.6220. The average of these two threshold levels is 0.4977 $((0.3734 + 0.6220)/2)$, which is very close to the critical value of 0.5318 calculated based on Regression (2.9).

To further examine how the effect of the intensity of chemical fertilizer application on urban-rural income disparity depends on the level of urbanization and whether this effect changes with time, by following Zhong (2011), we remove the interaction term in Eq. (1) and reintroduce the new interaction terms constructed by multiplying the intensity of chemical fertilizer application, urbanization level, and the year dummy. Next, we estimate the following regression equation:

$$gap_{it} = \gamma_0 + \gamma_1 L \cdot gap_{it} + \gamma_2 fer_{it} + \gamma_3 urban_{it} + \sum_t \theta_t \times (crossyear_{it}) + \sum_j \alpha_j \times control_{jt} + u_i + y_t + e_{it} \quad (3)$$

where $crossyear_{it}$ is the new interaction item, and θ_t is the coefficient of the new interaction. Eq. (3) allows us to examine whether the coefficients and the statistical significance of the interaction terms between the intensity of chemical fertilizer application and the level of urbanization will change over time.

In Table 7, the results indicate that the effects of the intensity of fertilizer application on narrowing urban-rural income disparity significantly depend on the level of urbanization, and such effects are significant in different years. However, the point estimates of interaction terms vary in size: the largest coefficient is 6.82 (*cross1998*) and the smallest coefficient is 3.67 (*cross2013*). Thus, the aforementioned threshold effects of urbanization show differences in different years; therefore, the corresponding regime-switching values are not stable. This time-varying characteristic of threshold levels of urbanization explains why the estimated threshold effect in Table 5 is only significant at the 10% level. The average of all estimated coefficients of the interaction terms in Table 6 is 4.97, which is close to the coefficient (4.16) of the interaction term in Regression (2.9) in Table 2.

We have offered evidence that the effect of the intensity of fertilizer application on urban-rural income disparity depends on urbanization. In addition to chemical fertilizers, other agricultural inputs, such as pesticide, agricultural machinery, agricultural plastic film, agricultural water, and rural electricity consumption, all of which may widen or narrow urban-rural income disparity, may be affected by urbanization. For example, the use of irrigation systems can effectively reduce rural poverty (Bhattarai et al., 2002). However, their impacts on poverty alleviation may depend on whether rural surplus labor forces that are released by the efficiency-improved irrigation systems from lands can be employed in non-agricultural sectors. Will our results change after considering the interaction terms between these five variables and urbanization? To test this prediction, based on Eq. (1), we introduce five extra interaction terms—*irrX*, *eleX*, *pestX*, *filmX*, and *powerX*—and use the system GMM estimator to re-estimate Regression (2.9).

In Table 8, the results indicate that the estimated coefficients of the aforementioned new interaction terms are not statistically

Table 6

Estimation results of the threshold effect considering non-overlapping 6-year averages.

Independent variable	Coefficient	Independent variable	Coefficient	Independent variable	Coefficient	Independent variable	Coefficient
<i>Lurban</i>	−2.2506 (1.5731)	<i>Lfdi</i>	−0.2292 (4.4295)	<i>Lfilm</i>	1.6748 (1.1575)	<i>Lfer*I (Lurban ≤ 0.6220)</i>	−4.7351*** (1.3621)
<i>Lrgdp</i>	−0.3277 (0.6298)	<i>Lgov</i>	−7.1859*** (2.3698)	<i>Lpower</i>	2.8309 (8.0674)	<i>Lfer*I (Lurban > 0.6220)</i>	0.2010 (1.1065)
<i>Linv</i>	0.2620 (0.5511)	<i>Lfd</i>	−0.3362 (0.2996)	<i>Lirr</i>	0.4106 (1.9206)	Threshold value	0.6220
<i>Lst</i>	0.2628 (0.3060)	<i>Ledu</i>	0.0101 (0.1756)	<i>Lele</i>	0.0823 (0.4587)	F test [p-value]	39.24 [0.0700]
<i>Lopen</i>	1.0053 (1.1782)	<i>Lpest</i>	4.9158*** (17.2452)	Constant	5.1138*** (0.9636)	Obs. (group)	60 (30)
						R ² (within)	0.8970

Notes: (1) Dependent variable: urban-rural income disparity; (2) the numbers in parentheses are robust standard errors; the numbers in square brackets are *p*-values; (3) ***, **, and * mean significant at 1%, 5%, and 10%, respectively.

Table 7
Time effect of the interaction terms.

Independent variable	Coefficient	Independent variable	Coefficient	Independent variable	Coefficient	Independent variable	Coefficient
<i>Lgap</i>	0.9021*** (0.0598)	<i>cross2003</i>	5.1765*** (1.9769)	<i>cross2011</i>	4.6840*** (1.7506)	<i>open</i>	−0.1812 (0.1111)
<i>fer</i>	−2.2401** (0.9982)	<i>cross2004</i>	5.5254*** (1.9532)	<i>cross2012</i>	4.1288** (1.7555)	<i>fdi</i>	0.2529 (0.4160)
<i>urban</i>	−2.0754** (0.8210)	<i>cross2005</i>	5.1770*** (1.9922)	<i>cross2013</i>	3.6744** (1.7591)	<i>gov</i>	−0.0703 (0.1986)
<i>cross1998</i>	6.8213*** (2.4687)	<i>cross2006</i>	4.7537** (1.9262)	<i>cross2014</i>	4.1207** (1.6095)	<i>fd</i>	0.0785 (0.0478)
<i>cross1999</i>	6.3104*** (2.2490)	<i>cross2007</i>	4.7554** (1.8882)	<i>cross2015</i>	4.1283** (1.6317)	<i>edu</i>	−0.1001*** (0.0375)
<i>cross2000</i>	5.9349*** (2.2619)	<i>cross2008</i>	5.0113*** (1.8918)	<i>rgdp</i>	0.0753 (0.0585)	<i>pest</i>	1.1365 (1.2061)
<i>cross2001</i>	5.3049** (2.1064)	<i>cross2009</i>	4.2469** (1.8258)	<i>inv</i>	−0.0228 (0.0846)	<i>film</i>	0.0952 (0.1090)
<i>cross2002</i>	5.0185** (2.1531)	<i>cross2010</i>	4.7617*** (1.7839)	<i>st</i>	0.0782** (0.0388)	<i>power</i>	−1.2436** (0.4897)
							Sargan test
							6.06 [0.300]

Notes: (1) Dependent variable: urban-rural income disparity; (2) the numbers in parentheses are robust standard errors; the numbers in square brackets are *p*-values; (3) ***, **, and * mean significant at 1%, 5%, and 10%, respectively; (4) the coefficients of *Lgap* estimated using the ordinary least-squares and the fixed-effect model are 0.9304 and 0.7982 respectively; (5) all of the regressions control for both the individual-specific effect and time-specific effect.

Table 8
Further checks on the effects of other agricultural inputs on urban-rural income disparity.

Indep.variables	(7.1) irr	Indep.variables	(7.2) ele	Indep.variables	(7.3) pest	Indep.variables	(7.4) film	Indep.variables	(7.5) power
OLS	0.9267	OLS	0.9286	OLS	0.9225	OLS	0.9278	OLS	0.9269
FE	0.7960	FE	0.7952	FE	0.7937	FE	0.7963	FE	0.7975
Lgap	0.8676*** (0.0978)	Lgap	0.8902*** (0.0553)	Lgap	0.9132*** (0.0700)	Lgap	0.8628*** (0.0724)	Lgap	0.8904*** (0.0641)
fer	-2.0551*** (0.6446)	fer	-2.0959*** (0.6569)	fer	-2.0959*** (0.6569)	fer	-2.4768*** (0.9336)	fer	-2.2582*** (0.6450)
urban	-2.2213** (0.8629)	urban	-2.0125*** (0.6759)	urban	-2.0125*** (0.6759)	urban	-2.4654** (0.9928)	urban	-2.0099*** (0.5920)
cross	4.0341*** (1.1538)	cross	4.0442*** (1.1681)	cross	2.8107*** (0.9520)	cross	4.6986*** (1.6234)	cross	4.3451*** (1.0957)
rgdp	0.0853 (0.0808)	rgdp	0.0368 (0.0523)	rgdp	0.0396 (0.0434)	rgdp	0.0317 (0.0762)	rgdp	0.0523 (0.0607)
inv	0.0091 (0.0840)	inv	0.0935 (0.1007)	inv	-0.0388 (0.0901)	inv	0.0282 (0.0987)	inv	0.0471 (0.0821)
st	0.0591* (0.0333)	st	0.0672* (0.0360)	st	0.0443 (0.0323)	st	0.0653* (0.0381)	st	0.0484 (0.0308)
open	-0.0022 (0.0888)	open	0.0427 (0.1036)	open	0.0116 (0.0828)	open	0.0249 (0.1302)	open	0.0104 (0.0808)
fdi	0.6373 (0.5829)	fdi	0.6254 (0.4987)	fdi	0.8044 (0.6639)	fdi	1.0083 (0.6951)	fdi	0.7365 (0.5842)
gov	0.1213 (0.2006)	gov	-0.0029 (0.1921)	gov	0.1077 (0.1627)	gov	0.0983 (0.1978)	gov	0.0405 (0.1718)
fd	0.0406 (0.0679)	fd	0.0135 (0.0559)	fd	-0.0088 (0.0505)	fd	-0.0059 (0.0600)	fd	0.0092 (0.0540)
edu	0.0007 (0.0437)	edu	-0.0241 (0.0321)	edu	-0.0182 (0.0300)	edu	-0.0035 (0.0380)	edu	-0.0181 (0.0324)
pest	0.9585 (0.9649)	pest	1.0741 (0.9991)	pest	-14.5233 (10.9144)	pest	0.7516 (1.5213)	pest	0.7583 (0.7796)
film	0.1170 (0.1452)	film	0.0774 (0.1088)	pestX	29.2188 (19.9264)	film	-0.2493 (0.3018)	film	0.0916 (0.1097)
power	-0.7256 (0.5056)	power	-0.6376* (0.3854)	film	0.0261 (0.0930)	filmX	0.6786 (0.6474)	power	0.0358 (1.5541)
irr	1.1635 (1.5245)	irr	0.0062 (0.1224)	power	-0.2771 (0.3388)	power	-0.8145 (0.5094)	powerX	-1.4056 (2.7786)
irrX	-0.5563 (0.7323)	ele	-0.3364 (0.3463)	irr	-0.0235 (0.1252)	irr	-0.0821 (0.1444)	irr	-0.0496 (0.1203)

(continued on next page)

Table 8 (continued)

Indep.variables	(7.1) <i>irr</i>	Indep.variables	(7.2) <i>ele</i>	Indep.variables	(7.3) <i>pest</i>	Indep.variables	(7.4) <i>film</i>	Indep.variables	(7.5) <i>power</i>
<i>ele</i>	−0.0177 (0.0666)	<i>eleX</i>	0.5414 (0.4990)	<i>ele</i>	−0.0117 (0.0468)	<i>ele</i>	0.0391 (0.0638)	<i>ele</i>	0.0146 (0.0470)
Constant	1.2275** (0.5797)	Constant	1.1479*** (0.4254)	Constant	1.0841** (0.5474)	Constant	1.5132** (0.6821)	Constant	1.1931*** (0.4566)
AR (1)	−3.56 [0.000]	AR (1)	−4.02 [0.000]	AR (1)	−3.97 [0.000]	AR (1)	−4.25 [0.000]	AR (1)	−3.95 [0.000]
AR (2)	0.27 [0.791]	AR (2)	0.35 [0.729]	AR (2)	0.18 [0.860]	AR (2)	0.31 [0.755]	AR (2)	0.26 [0.795]
Sargan	6.17 [0.187]	Sargan	7.13 [0.211]	Sargan	6.86 [0.231]	Sargan	7.16 [0.209]	Sargan	7.74 [0.171]
test		test		test		test		test	

Notes: (1) Dependent variable: urban-rural income disparity; (2) the numbers in parentheses are robust standard errors; the numbers in square brackets are p-values; (3) ***, **, and * mean significant at 1%, 5%, and 10%, respectively; (4) OLS and FE represent the coefficients of *Lgap* estimated using the ordinary least-squares and the fixed-effect model, respectively; (5) observations are 540, and groups are 30; (6) all of the regression models control for both the individual-specific effect and time-specific effect; (7) the variables *irrX*, *eleX*, *pestX*, *filmX*, and *powerX* represent the interaction terms of corresponding variables with the urbanization.

significant. Additionally, the estimated coefficients and statistical significance of the intensity of chemical fertilizer application, urbanization, and their intersection term are not affected by adding these five new interaction terms into the regressions. This result further confirms the robustness of the regressions.

6. Concluding remarks

At present, China is confronted with an extremely complicated international trade environment and considerable deceleration risks in the process of economic development (Eichengreen, Park, & Shin, 2012; Hong, Lee, Liao, & Seneviratne, 2017; Huang & Tian, 2016; Hughes, 2005). Therefore, the focus of improving China's economic development efficiency should be on the reforms of China's economic structure and corresponding social systems. Undoubtedly, reducing such risks and improving economic efficiency are closely related to curbing the continuously widening income gap, achieving a balanced development between urban and rural areas, and implementing the *Rural Revitalization Strategy* proposed by the central government of China. Under such backgrounds, this paper explores the way to mitigate China's urban-rural income disparity from the perspectives of agricultural inputs, urbanization, and their interaction.

In this paper, we offer empirical evidence that the intensity of chemical fertilizer application has a significant impact on urban-rural income disparity, but this effect depends on the level of urbanization. We also draw the following main conclusions. First, urbanization has a statistically significant effect on mitigating urban-rural income disparity. Second, at lower levels of urbanization, increasing the intensity of chemical fertilizer application can significantly moderate urban-rural income disparity; at higher levels of urbanization, reducing the intensity of chemical fertilizer application is conducive to narrowing urban-rural income disparity. Third, the mechanism of regime switching, which depends on the levels of urbanization, shows some time-varying characteristics; however, the threshold effects are statistically significant over the entire sample period. Fourth, for other agricultural inputs, such as pesticides and agricultural plastic films, similar threshold effects have not been observed. Although the impact of the intensity of chemical fertilizer application on urban-rural income disparity and the threshold effect of urbanization have not been empirically tested in existing literature, the research on this issue has important practical significance.

Because of the strong dependence of China's agricultural production on the use of chemical fertilizers and the adverse effects caused from excessive chemical fertilizer application, such as water, air, and land pollution and damages to farmers' health, more stringent control over the amount and intensity of chemical fertilizer application in agricultural production is necessary and urgent. However, as the increased chemical fertilizer application has had a significant impact on narrowing the urban-rural income disparity in China, relevant fertilizer application control policies, such as the *Action Plan of Zero Growth in Fertilizer and Pesticide Use by 2020* issued by the MOA, may have negative effects on mitigating urban-rural income disparity in the short term, which may in turn result in the conflicts between environmental regulation policies and income promotion policies. More importantly, in the face of such policy conflicts, it is difficult to balance economic efficiency and social equity.

The urbanization level in China remains relatively low, and most cities can still absorb a large number of rural immigrations. The urbanization per se is a key force in narrowing the urban-rural income disparity in China. Moreover, urbanization indirectly affects the combination of agricultural production inputs through the income growth effect. Hence, it can be expected that with the advancement of urbanization, the opportunities for farmers to pursue income growth will increase, and the dependence on chemical fertilizers and other agrochemicals will gradually decline. Thus, urbanization can play an important role in improving the allocation efficiency of agricultural production inputs, reducing the risks of environmental pollution, and eliminating urban-rural conflicts. In particular, to fulfill the residents' pursuit of a healthier life, the environmental protection and governance in China should be intensified; meanwhile, the effect of urbanization on improving environmental quality should be emphasized.

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